

JIN ZIYU

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Summary

BEng (Hons) Robotics Systems Undergraduate Driven Robotics Systems undergrad with hands-on expertise in electronics, automation, and quality engineering. Adept at robotics inspection and precision measurement, eager to contribute to innovative teams and advance manufacturing automation.

Education

Temasek Polytechnic	Singapore, Singapore
Diploma in Electronics	03/2019 - 04/2022
- Acquired extensive knowledge of electronic circuits, systems, and components. - Honed expertise in design, diagnostics, and maintenance of electronic devices and systems. - Emphasized practical application and adherence to industry standards. - Achieved commendation award in Major project given to 10% of the number of projects in year 3, semester I - Relevant coursework: Data Visualization and Analytics, Smart Manufacturing System, Robotics and Automation, Circuits and Control Systems, PCB Design, Circuit Analysis, Analog Electronics	
Temasek Polytechnic	Singapore, Singapore
Specialist Diploma in Robotics & Automation	10/2022 - 09/2023
- Advanced proficiency in robotics systems and automation technologies. - Specialized in control systems for robotics applications. - Expertise in designing and programming robotic solutions. - Optimized manufacturing processes through automation. - Enhanced operational efficiency with cutting-edge robotics. - Relevant coursework: Manufacturing Equipment & Troubleshooting, Automation Control & Interfacing, Robotics Systems, control & Troubleshooting, Robotics Integration & Adaption	
Singapore Institute of Technology	Singapore, Singapore
BEng (Hons) Robotics Systems	08/2024 - Present
Relevant Coursework: C++ Programming, Embedded Systems, Analog and Digital Electronics, System Engineering Project I (ROS1), Project 3 (ROS2), Systems Engineering	

Experience

Kinergy Corporation Ltd	Location
Assistant quality assurance engineer	05/2022 - 07/2024
Company Description	
- Conducted thorough equipment inspections utilizing UV light technology to ensure adherence to quality standards - Employed precision measurement tools such as feeler gauges to maintain equipment accuracy and performance - Demonstrated expertise in electronics and robotics, contributing to the enhancement of quality assurance processes - Developed solutions to technical challenges, optimizing process efficiency and product reliability - Collaborated with the quality assurance team to identify areas for process improvement and implement strategic changes - Maintained a strong focus on quality control, ensuring all products met rigorous testing and inspection criteria	
Wizlogix Pte Ltd	Location
Supply chain intern	10/2021 - 03/2022
Company Description	
- Monitored inventory levels to ensure accuracy and promptly addressed discrepancies - Implemented problem-solving strategies to resolve emergency issues effectively - Generated Purchase Orders and Goods Received Notes within SAP (Systems Applications and Products in Data Processing) to maintain procurement workflow - Collaborated with a team of 6 colleagues in the supply chain department to optimize operations and support departmental goals	

Projects

Hand Gesture Controlled Robot	04/2021 - 08/2021
- Constructed, assembled, and programmed a six-axis robot for efficient pick-and-place operations, leveraging 3D printing and Arduino nano microcontroller technology to optimize movement across each axis. - Established seamless wireless communication between the robot and a Haptic Glove by configuring HC05 Bluetooth modules in a Master-Slave configuration. - Spearheaded a team of three members, overseeing meeting arrangements and task delegation to ensure the timely completion of project objectives, and received a certificate of commendation in recognition of achievement	
LIMO Arena Design & Navigation	04/2025 - 07/2025
- Engineered an autonomous navigation arena for the LIMO robot platform using the SEBoK Systems Engineering methodology. - Designed realistic and modular arena environments optimized for robot localization, mapping, and path-planning performance. - Conducted iterative testing and validation to improve navigation reliability, obstacle avoidance, and operational efficiency. - Coordinated project planning, requirement analysis, and subsystem integration within a collaborative engineering team.	

Projects

Intelligent Pallet & Optimization System	01/2025 - 04/2025
• Designed and developed an autonomous pallet measurement system integrating load cells and a 3D stereo vision camera to capture weight and dimensional data with minimal human intervention. • Implemented a ROS 2–based system architecture with TCP communication to enable real-time data acquisition, synchronization, and processing between hardware and software components. • Engineered a modular mechanical structure using aluminum profiles and 3D-printed components to ensure stability, accurate load distribution, and scalability. • Conducted system integration, testing, and validation to improve measurement accuracy, data synchronization, and overall system reliability for warehouse operations.	

Leadership and Co-Curricular Experience

SIT- Project Hub	
Student Coach	03/2025 - 03/2026
• Mentored students in the use of makerspace tools, including 3D printing, laser cutting, and microcontrollers (Arduino, Raspberry Pi), supporting design, prototyping, and fabrication processes. • Guided peers in troubleshooting technical issues, promoting innovative thinking and collaborative learning. • Ensured adherence to lab safety protocols and operational procedures, maintaining a safe and organized working environment. • Assisted in managing makerspace activities, supporting both independent and team-based projects while improving user competency with engineering tools.	
SIT- First Year Experience: Day Zero	
Facilitators	08/2025 - 08/2025
• Assisted in facilitating freshman orientation activities, supporting engagement and smooth execution of programme sessions. • Helped manage logistics, including attendance coordination, classroom setup, and distribution of materials and meals. • Guided freshmen between venues and ensured proper crowd movement during large-scale event transitions. • Worked with the organizing team to address on-ground issues and maintain an efficient and positive event experience.	

Skills

Software Skills: C/C++, Language PLC programming, Arduino, Python Programming, ROS 1, ROS 2 Humble

IT Skills: Proficient in Microsoft Word, Excel, PowerPoint, Teams, Zoom, SolidWorks, EAGLE, Tableau

Language: Fluent in English and Chinese